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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS 1-1 **Date: 09/16/2026**

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If

you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you do not.

Part E – Real-World Job Search (Optional / Bonus)

7. Search for a real job platform (e.g., LinkedIn, indeed, Job Street, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, testing, and building applications and algorithm-driven software systems.	Software Engineer, Backend Developer, Full-Stack Engineer.
Systems & Cloud Infrastructure	Maintaining hardware provisioning, network virtualization, uptime, and deployment pipelines.	Systems Administrator, DevOps Engineer, Site Reliability Engineer (SRE).
Data Science & AI Engineering	Extracting insights from structured/unstructured data and training predictive AI models.	Data Scientist, Machine Learning Engineer, Data Engineer.
Cybersecurity & Threat Defense	Safeguarding network perimeters, conducting penetration testing, and managing incident response.	Information Security Analyst, Penetration Tester, SOC Specialist.
Information Systems & Enterprise IT	Aligning business workflows with technology solutions and enterprise software architectures.	Business Systems Analyst, IT Manager, Solution Architect.

My Chosen Pathway (Part B)

I am most interested in: Software Engineering

Why I chose it:

Software Engineering can be interesting on how fast technology progresses. Software Engineering mostly creates and plans software architecture, it uses clean code using programming that we are studying in our program. I find coding a puzzle-solving problem, it's somewhat addicting to see how it is structured. Seeing there's no end in learning new things in coding; I find it really fun on the pursuit of new knowledge without limit.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Junior Coder	Software Engineering Intern, Junior Software Engineer, Associate Software Engineer.	Writing clean basic code, Debugging and fixing errors, taking part in code reviews, Learning frameworks
Mid-Level Developer	Software Engineer, Member of Technical Staff (MST)	Independent feature ownership, code quality & reviews, Unit Testing, Code reviews.
Senior Engineer	Senior Software Engineer, Engineering Manager/Tech Lead, Staff Engineer.	System Architecture Design, Scalability and Performance, Mentoring Juniors, Full stack Responsibility.
Technical Lead	Tech Lead	Guiding technical decisions, Team leader, Architecture Overviews, Cross-functional Collaboration.
Software Architect	Solution Architect, Enterprise Architect, Data/Infrastructure Architect	High-level System Design, Defining Standards, Integrating Complex Systems, Full Lifecycle Strategy

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Data Structure	Y	
Software Architecture	N	Research, books that are available online and any videos that explain what it is.
Object-oriented Design	N	Research, books that are available online and any videos that explain what it is.
CI/CD	N	Research, books that are available online and any videos that explain what it is.
Git	N	Research, books that are available online and any videos that explain what it is.

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Video Game Designer
Company	N/A
Platform / Source (e.g., LinkedIn, indeed, Job Street)	LinkedIn

Item	Details
Required Skills Listed (3–5)	Game Design Documents (GDD), Level Design & Architecture, Game Mechanics & Systems Design, Scripting & Blueprinting, Coding Languages, Game Engines.

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I really want this pathway because it's the closest to be a video game designer, I had a friend who promised me that we were going to make a visual novel (A game novel like) and I was responsible for design on the art parts. And with learning this path, I could do the coding too and help both our dreams finally come true.

2. Looking at your roadmap, which career stage excites you the most, and why?

The most exciting thing for any job is just starting and learning, so I chose Junior Coder. I think this would be my motivation to create something great for everyone who wants to see and experience our creation. It would be my starting point and my pride if I ever get the job.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

I know game designs and artworks on game structures work since I did it before on most games. The hardest thing to learn would possibly be complex coding and trying to debug every bug I meet, despite being hard it would be interesting and motivating for me to fight over.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

It would help me to set the right path and what to expect on my journey ahead. It would easier to plan a roadmap so I won't get lost or keep stepping down on this ladder of life.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25

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Criteria	Description	Points
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheets are legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date